NOTRE DAME UNIVERSITY
BANGLADESH

Lab Task - 05

Course Title: Technical Writing & Presentation Lab

Course Code: CSE 3207

Submitted by:

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Assignment Topic: Typesetting Exercise

Submitted to:

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Lab Task-05

Typesetting Exercise 1 :

Code in Texmaker :

```
\documentclass[13pt]{article}
\usepackage{enumitem}
\usepackage{amsmath}
\begin{document}
\title{Practice Lab-05 : \\ Packages}
\author{Istiak Alam}
\maketitle

\begin{enumerate}[label=\Alph*.]
    \item 1/3 cup and 2 and 1/2 tablespoon chicken
    \item 1/2 medium tomato
    \item Red chilli powder as required
    \item 1 tablespoon vegetable oil
    \item 1/2 tablespoon green chilli sauce \\ \\
\end{enumerate}
With Equation Number
\begin{equation}
\Omega = \sum_{k=1}^{n} \omega_k
\end{equation}
Without Equation Number
\begin{equation*}
\Omega = \sum_{k=1}^{n} \omega_k \\ \\
\end{equation*}
\begin{equation*}  %bad
min_{x,y} (1-x)^2 + 100(y-x^2)^2
\end{equation*}
\begin{equation*}  %Good
\min_{x,y}{(1-x)^2+100(y-x^2)^2}
\end{equation*}
\begin{equation*}
\beta_i =
\frac{\operatorname{Cov}(R_i, R_m)}
         {\operatorname{Var}(R_m)}
\end{equation*}

\begin{align*}
(x+1)^3 &= (x+1)(x+1)(x+1) \\
                &= (x+1) (x^2 + 2x + 1) \\
                &= x^3 + 3x^2 + 3x + 1
\end{align*}
\end{document}
```

Output in Texmaker :

Practice Lab-05 : Packages

Istiak Alam

April 9, 2025

A. 1/3 cup and 2 and 1/2 tablespoon chicken

B. 1/2 medium tomato

C. Red chilli powder as required

D. 1 tablespoon vegetable oil

E. 1/2 tablespoon green chilli sauce

With Equation Number

$$\Omega = \sum_{k=1}^{n} \omega_k \tag{1}$$

Without Equation Number

$$\Omega = \sum_{k=1}^{n} \omega_k$$

$$min_{x,y}(1-x)^2 + 100(y-x^2)^2$$

$$\min_{x,y} (1-x)^2 + 100(y-x^2)^2$$

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}$$

$$\begin{aligned}(x+1)^3 &= (x+1)(x+1)(x+1) \\ &= (x+1)(x^2+2x+1) \\ &= x^3+3x^2+3x+1\end{aligned}$$

Typesetting Exercise 2 :

Code in Texmaker :

```
\documentclass[13pt]{article}
\usepackage{amsmath}
\begin{document}
\title{Lab Task-05 : \\ Typesetting Exercise 2}
\author{Istiak Alam}
\maketitle

\section{Typeset in LaTex}
%\textbf{Typeset in LaTex}
Let $X_1,X_2,...,X_n$ be a sequence of independent and identically
distributed random variables with E$[X_i]=\mu$ and Var$
[X_i]=\sigma^2<\infty$ \\

\begin{equation*}
S_n = \frac{1}{n} \sum^n_i X_i
\end{equation*}

donote their mean. then as $n$ approaches infinity, the random variables $
\sqrt{n}(S_n-\mu)$ converge in distribution to a normal $N(0,\sigma^2)$.

\end{document}
```

Output in Texmaker :

Lab Task-05 :
Typesetting Exercise 2

Istiak Alam

April 9, 2025

1 Typeset in LaTex

Let $X_1, X_2, ..., X_n$ be a sequence of independent and identically distributed random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$

$$S_n = \frac{1}{n} \sum^n_i X_i$$

donote their mean. then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $N(0, \sigma^2)$.

Figure 1: Output in PDF